

Week 12 – Documentation & Future Scope Planning

Summary:

Week 12 focused on **creating comprehensive documentation** of backend functionality and planning the **future frontend integration**.

Key Activities:

- Documented all backend endpoints, Docker setup, multi-language support, and library testing.
- Prepared **weekly and cumulative reports** in PDF format.
- Proposed a **frontend interface** for future web-based code submission and visualization.
- Reviewed backend performance logs and prepared notes for any future improvements.

Technologies Used: Python, FastAPI, Docker, Markdown/PDF, WSL2

Outcome:

Backend is fully functional, well-documented, and ready for future **frontend implementation**. Documentation ensures reproducibility and ease of understanding.

Flowcharts:

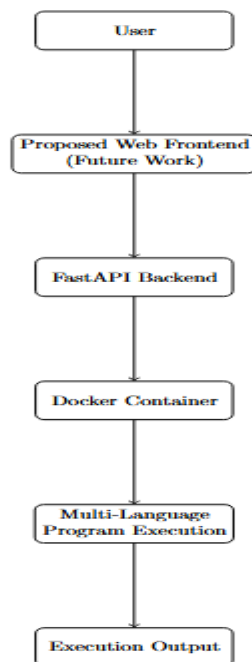


Figure 1: Proposed Frontend-Backend Integration Flow (Future Scope)